

QUESTION -

1. Stock Buy and Sell - Max one Transaction Allowed

Given an array `prices[]` of non-negative integers, representing the prices of the stocks on different days, find the maximum profit possible by buying and selling the stocks on different days when at most one transaction is allowed. Here one transaction means 1 buy + 1 Sell. If it is not possible to make a profit then return 0.

Examples:

Input: prices[] = [7, 10, 1, 3, 6, 9, 2]

Output: 8

Explanation: Buy for price 1 and sell for price 9.

Test Case	Input String	Expected Output
1	[7, 10, 1, 3, 6, 9, 2]	8
2	[7, 6, 4, 3, 1]	0
3	[1, 3, 6, 9, 11]	10
4	[2, 3, 6, 9, 11]	9
5	[7, 6, 4, 3, 2]	0

SOLUTION -

```
class Solution:
    def maxProfit(self, prices):

        # Lowest buying price seen so far
        min_price = prices[0]

        # Maximum profit found so far
        max_profit = 0

        # Traverse all prices
        for price in prices:

            # Found a cheaper buying price
            if price < min_price:
                min_price = price

            # Calculate today's profit
            profit = price - min_price

            # Update maximum profit
            if profit > max_profit:
                max_profit = profit

        return max_profit
```